

# Project P.U.L.S.E.

Predictive Unified Life-sciences Summarization Engine

Model Report — Generated March 27, 2026

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## 1. Executive Summary

This report summarizes the model development, data quality assessment, and drift monitoring results for the P.U.L.S.E. Clinical Intelligence Platform, built to support the Abbott Global Data Science & Analytics (GDSA) team. The platform integrates five clinical datasets, two predictive models, and a Real-World Evidence (RWE) drift monitoring system.

## 2. Dataset Overview

| Dataset      | Patients | Labeled (DM) | Avg Age | Avg BMI |
|--------------|----------|--------------|---------|---------|
| chronic      | 3498     | 3498         | 53.1    | 27.1    |
| diabetes_130 | 101766   | 101766       | 66.0    | 27.1    |
| nhanes       | 21904    | 13373        | 35.4    | 27.1    |
| statlog      | 270      | 0            | 54.4    | 27.1    |

## 3. Model Performance Summary

Both models were trained using XGBoost with 5-fold stratified cross-validation. SHAP values were computed for feature interpretability. All experiments are tracked in MLflow for reproducibility.

| Model              | Vertical       | CV AUC (mean) | Algorithm | Status     |
|--------------------|----------------|---------------|-----------|------------|
| DiabetesClassifier | Metabolic      | ~0.82         | XGBoost   | Production |
| CardioClassifier   | Cardiovascular | ~0.86         | XGBoost   | Production |

## 4. Real-World Evidence (RWE) Drift Monitoring

The Sentinel Suite compares the NHANES 2015-16 (pre-pandemic baseline) against the NHANES 2021-23 (post-pandemic production stream). Evidently AI is used to detect data drift and concept drift across all key biomarkers.

| Biomarker | Drift Detected | Action Recommended |
|-----------|----------------|--------------------|
|-----------|----------------|--------------------|

|                 |          |                           |
|-----------------|----------|---------------------------|
| Glucose (mg/dL) | Yes      | Retrain on post-2020 data |
| HbA1c (%)       | Yes      | Retrain on post-2020 data |
| BMI             | Moderate | Monitor quarterly         |
| Systolic BP     | No       | No action required        |

## 5. Recommended Next Steps

- Retrain DiabetesClassifier on pooled 2015-16 + 2021-23 NHANES data
- Expand cardiovascular vertical with additional datasets (e.g., MIMIC-III)
- Deploy Streamlit dashboard to internal Streamlit Cloud for team access
- Integrate PubMed abstract scraping into the RAG knowledge base
- Schedule monthly drift detection runs via GitHub Actions cron job